

# Bringing Work Home

Flexible Work Arrangements as Gateway Jobs for Women in West Bengal

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## Motivation

- Large, latent labor force in low- and middle-income countries (ILO and Gallup, 2016)
    - Hundreds of millions of women *prefer* to have paid work but are out of the labor force
    - Concentrated in South Asia, the Middle East, and North Africa
- India:**  $\approx 166$  million women want paid work but are out of the labor force

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- High stakes to getting women into the labor force
  - Better fulfill individual preferences
  - More efficient allocation of labor  $\rightarrow$  higher aggregate output (HHJK 2019, ABMQ 2023)

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- High stakes to getting women into the labor force
  - Better fulfill individual preferences
  - More efficient allocation of labor  $\rightarrow$  higher aggregate output (HHJK 2019, ABMQ 2023)
- Where's the bottleneck? In India, **labor market entry** is a major challenge
  - Only 29% of 30-year-old women in India have ever worked for pay (PLFS 2021-2022)
  - In comparison,  $\approx 90\%$  of men aged 25-29 say they've recently worked for pay

# Motivation

- Why don't these women work? Mismatch in jobs available vs jobs women can do given norms
  - The median married women in India spends 0.5 hours outside her home per day  
(Andrew and Smurra 2024)
- Mismatch suggests two potential strategies: (1) change norms or (2) change jobs
  - This project: **start by changing jobs** without directly addressing norms
  - Introduce digital gig work opportunities (emerging job amenable to working from home)
  - In turn, demonstration of women earning income may prompt more pro-work attitudes

## Randomized field experiment with 1,670 out-of-labor-force women in West Bengal

- 1. Labor market entry:** offer randomly selected households digital gig jobs for women that vary in flexibility along three dimensions: (1) time, (2) multitasking with childcare, (3) work-from-home  
*Result:* these jobs are highly effective at drawing out-of-labor-force women into paid work, increasing take up by **>3x** from **15%** (office) to **48%** (most flexible, work-from-home job)
- 2. Gateway jobs:** offer another set of short-term jobs 2-3 months later that vary in flexibility  
*Result:* home-based jobs act as a stepping stone to less flexible, outside-the-home work (+**8 pp** likelihood in starting the second job, **0.1 SD** shift towards less traditional attitudes)
- 3. Labor demand side:** estimate effects of flexibility on worker performance using randomly assigned surprise upgrades to job offers to separate selection versus treatment effects  
*Result:* Tradeoff to work from home — **3%** more accurate and **18%** faster from office

## Contribution to the literature

### 1. Flexible work arrangements and labor market gender gaps

- Most work has focused (a) on high-income countries and (b) on the intensive margin, showing women's higher WTP for job flexibility contributes to the gender wage gap (e.g. Goldin 2015; Mas & Pallais 2017; Wiswall & Zafar 2018)

*This project:* women's high valuation of job flexibility contributes to the gender gap in labor force participation (**extensive margin, first-order in countries like India**)

### 2. Effects of remote work on job performance and labor market attachment

### 3. The evolution of gender norms

### 4. Strategies to increase female labor force participation

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*This project*: (a) labor market *entry* in response to a work-from-home shock, (b) understudied population (secondary-school educated, simpler tasks, developing country), (c) novel dimension of flexibility—multitasking work with childcare

## 3. The evolution of gender norms

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## 3. The evolution of gender norms

- Gender attitudes & gendered labor supply are shaped by economic conditions (e.g. Boserup 1970; Fernández et al 2004; Alesina et al 2013; Field et al 2021)

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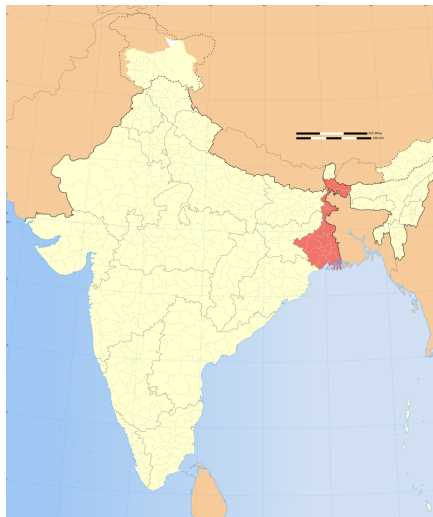
## 4. Strategies to increase female labor force participation

- Lots of recent work about determinants of women's labor supply (e.g. Bernhardt et al 2018; Heath & Jayachandran 2018; Burszstyn et al 2020; Field et al 2021; Lowe & McKelway 2023; McKelway 2023; Buchmann et al 2025; see Fletcher et al 2018 for a review)
- Even compared to effective interventions, **work-from-home** has a large effect on take up

# Roadmap

1. Introduction
2. Study Context
3. Result 1: large extensive margin labor supply response to job flexibility
4. Result 2: home-based jobs act as a stepping stone to less flexible, outside-the-home work
5. Result 3: tradeoff – workers produce output at lower quality, and more slowly, from home

# Study setting: urban, peri-urban, & rural West Bengal



Lower-middle-income HHs: Rs 11,177/month (136 USD)

To be eligible, woman could not be in labor force

- $>2/3$  had **never** worked for pay before study
- $\approx 1/3$  have the final decision in own labor supply

30 years old on average and 93% married:

- 40% live with father- or mother-in-law
- $3/4$  have a child under 18 years old

► More Demographics    ► Representativeness    ► Map of Recruitment Areas

## Prevailing gender norms confine many women to the home

**Many women have attitudes that adhere to stereotypical family gender roles:**

- 67% agree that woman's most important role is being good homemaker
- 64% agree that earning a living is father's responsibility, & childcare is mother's responsibility
- 52% agree that a man should have the final word about decisions in his home

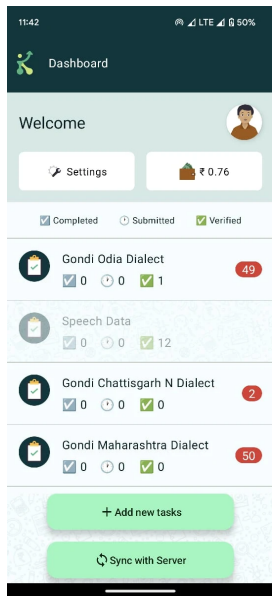
Aggregate responses to these (& 12 more) statements to create a **gender attitudes index**

- Classify as "more traditional" or "less traditional" in comparison to median participant

If women, husbands, in-laws have these views → difficult for women to take jobs outside the home

- We offer emerging type of work amenable to flexible work arrangements: **digital gig work**

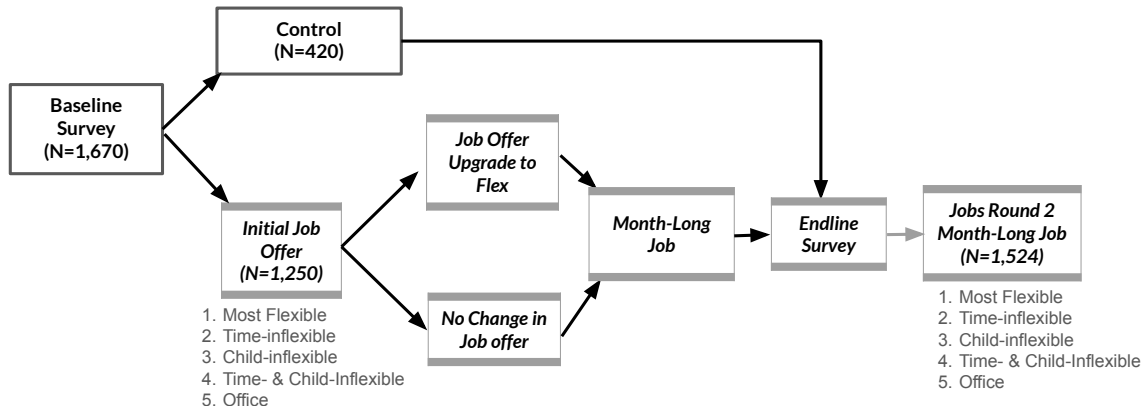
# Jobs implemented in collaboration with Microsoft Research / Karya



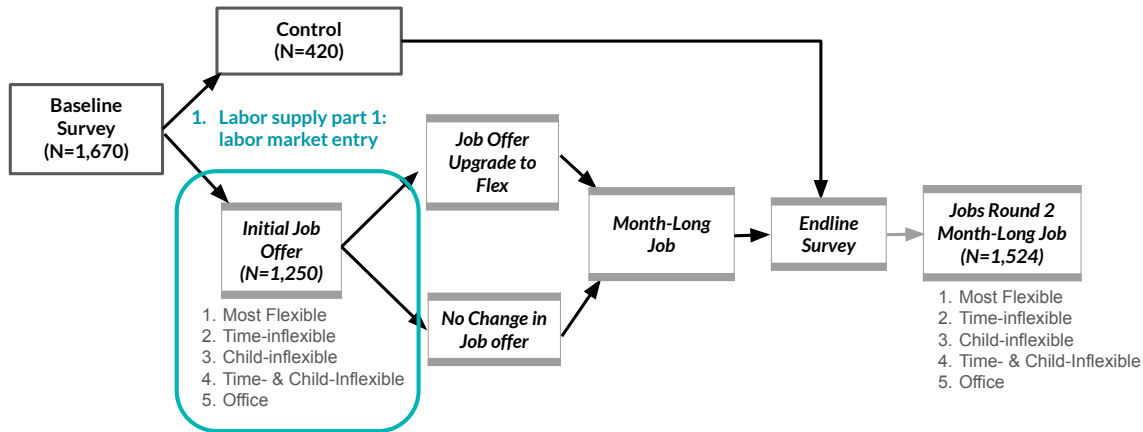
- Data annotation tasks to contribute to Bangla/Hindi datasets
  - **Real job tasks** clients requested & paid for
- Jobs lasted for one month, new tasks each week
  - Validators score tasks on quality (e.g. accuracy, noise)
- Pay: **piece-rate**, max. possible: Rs 4000 ( $\approx$  50 USD)
  - Average earnings: Rs 2500 (**21% of monthly HH income**)
  - Average work time: 22 hours over 14 days
  - Generous pay in comparison to available jobs

► Distribution: hours worked    ► Distribution: tasks completed    ► Distribution: earnings

# Design Overview

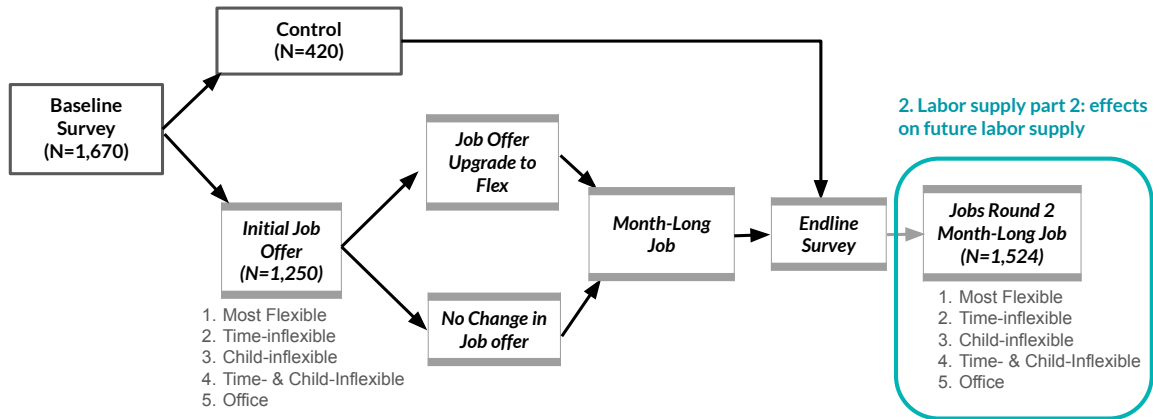


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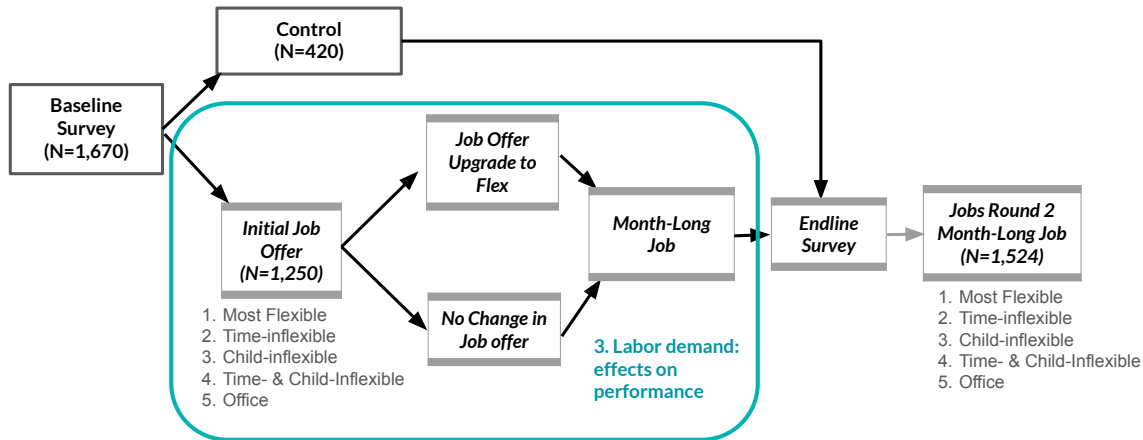




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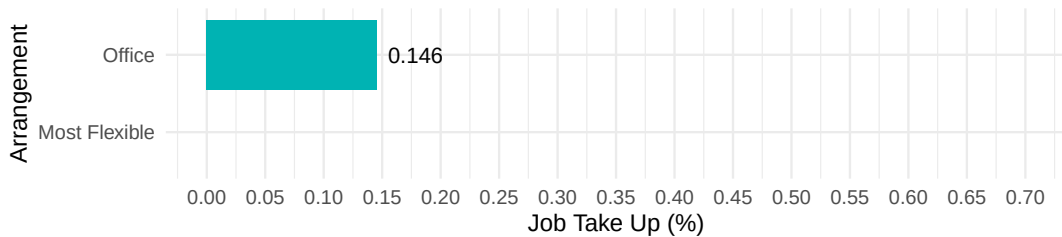
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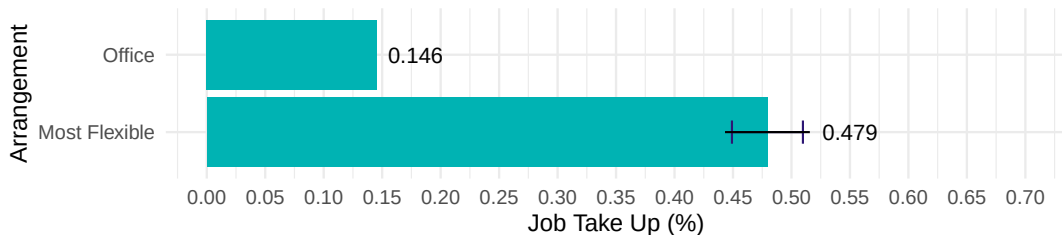
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## Result 1: large extensive margin labor supply response to job flexibility

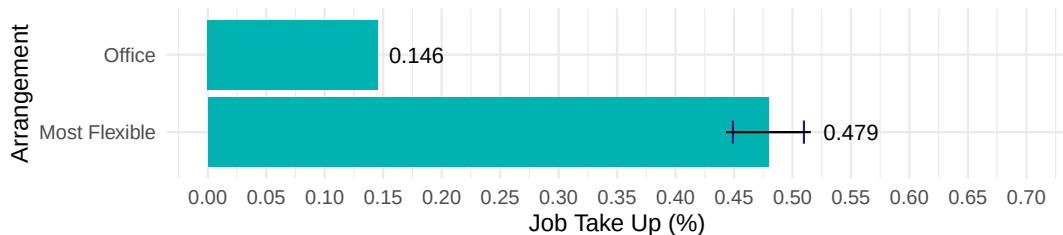


## Result 1: large extensive margin labor supply response to job flexibility



Bundled flexibility increases job take up **by more than 3x** (+33pp, 228% increase)

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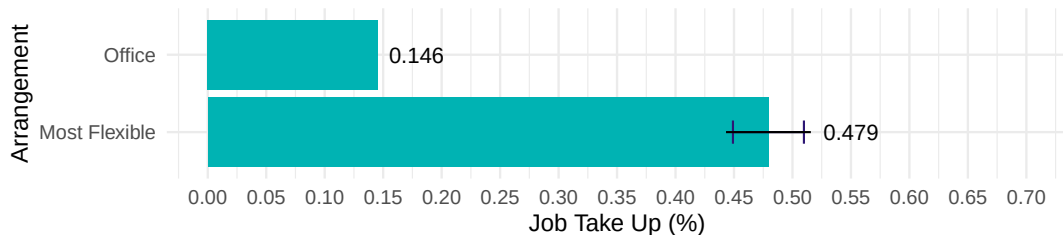


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This is a very large effect compared to other levers, e.g.

1. Correcting second order beliefs: +9pp (36%) sign up for job matching service (Bursztyn et al '20)
2. Self-efficacy intervention: +5pp (32%) (McKelway 2023)
3. Control over earnings: +5pp (13%) (Field et al 2021)
4. **≈5x wage increase from 5k to 24k INR: 7pp (32%)** (Jalota and Ho 2024)

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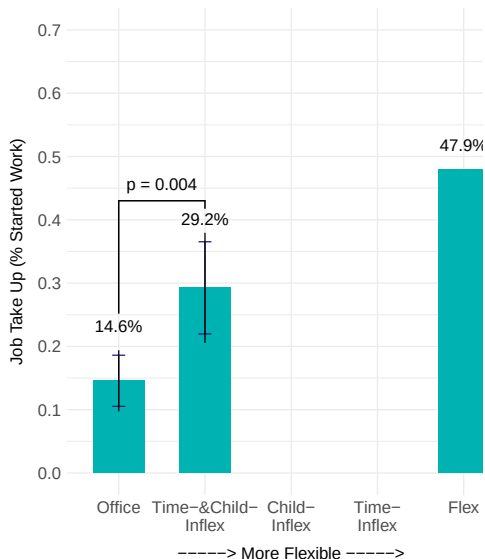
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**What mechanisms drive the importance of job flexibility for women's labor supply?**

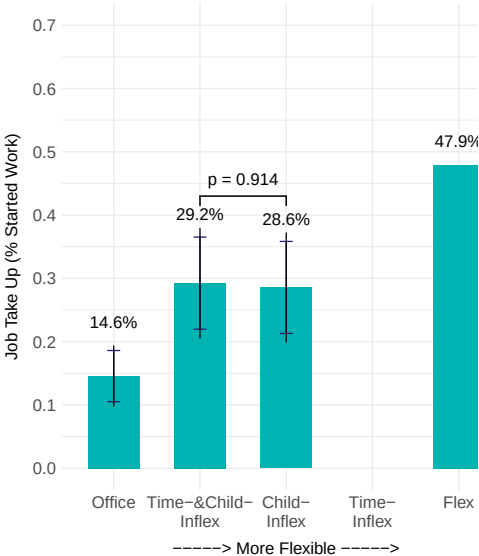
## Work-from-home (fixed hours, no multitasking with childcare) doubles job take up.



| Final Work Arrangement (N=1250)               | Choose Hours | Multitasking (Childcare) | Work from Home |
|-----------------------------------------------|--------------|--------------------------|----------------|
| 1. Most Flexible (N=731)                      | Yes          | Yes                      | Yes            |
| 2. Time-Inflexible (N=103)                    | No           | Yes                      | Yes            |
| 3. Child-Inflexible (N=105)                   | Yes          | No                       | Yes            |
| 4. Time-Inflexible & Child-Inflexible (N=105) | No           | No                       | Yes            |
| 5. Office (N=206)                             | (Yes)        | No                       | No             |

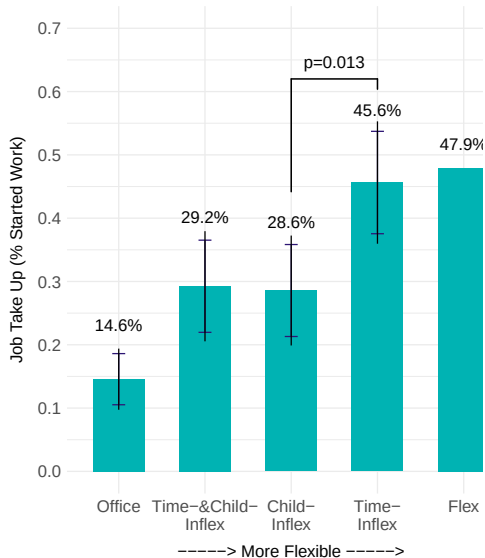


# Work hours flexibility on its own makes little difference to job take up.



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## Ability to multitask work with childcare increases job take up by 17 pp (59%)



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| 5. Office (N=206)                             | (Yes)        | No                       | No             |

## Takeaways: elasticity of labor supply with response to job flexibility

1. Flexible, work-from-home jobs are highly effective at increasing out-of-labor-force women's participation in paid work
  - The most flexible work-from-home job **more than triples** take up from **15%** to **48%**
2. Decomposing the 33 pp of women whose job take up is marginal to this flexibility:
  - $\approx$ half comes from the ability to **work from home** (even without child multitasking)
  - $\approx$ half comes from the ability to **multitask work with childcare**
  - $\approx$ no effect of time flexibility (conditional on choosing shift)

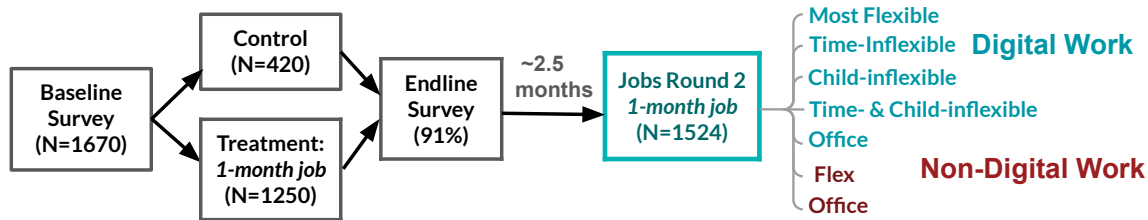
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## Effects of work experience on future labor supply

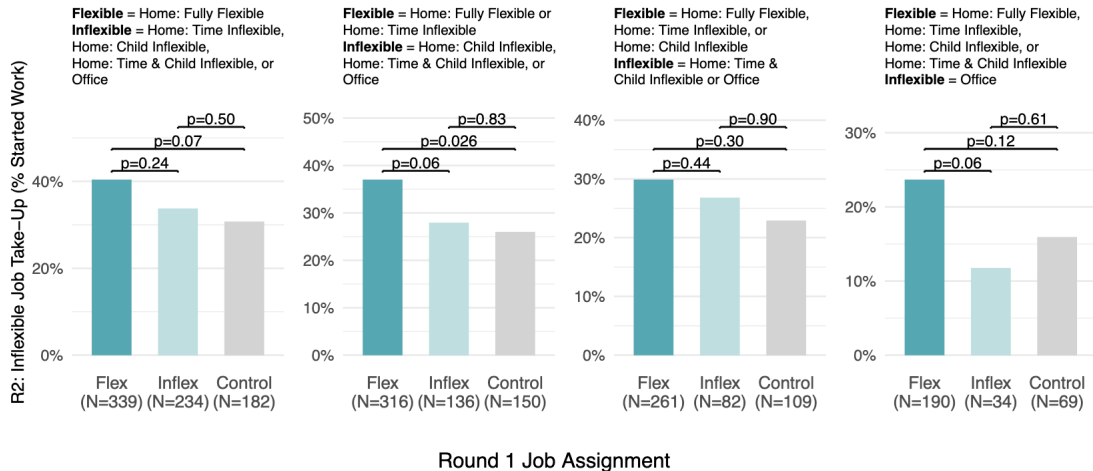
Many possible mechanisms, including:

1. Income effect: women who have eased household financial distress should be less likely to work in the second round of jobs
2. Demonstration of women earning income might change gender attitudes, lowering cost of taking up work in the future



# Take up of Inflexible Round 2 Jobs, by Flexibility of Round 1 Job Assignment

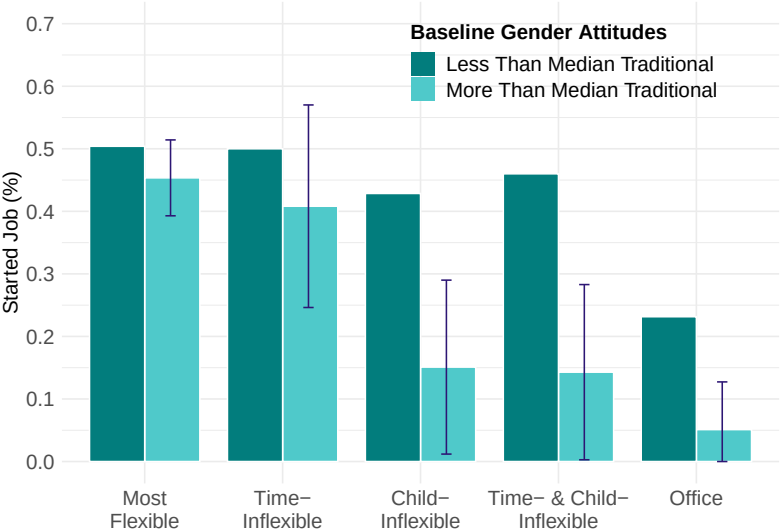
## First-Time Paid Workers: Inflexible R2 Take-Up



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  - Mechanism: work experience changes attitudes about appropriate behavior for women
5. Result 3: tradeoff – workers produce output at lower quality, and more slowly, from home

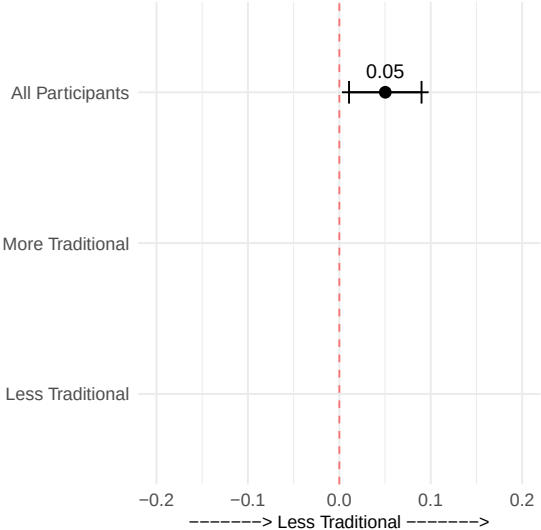
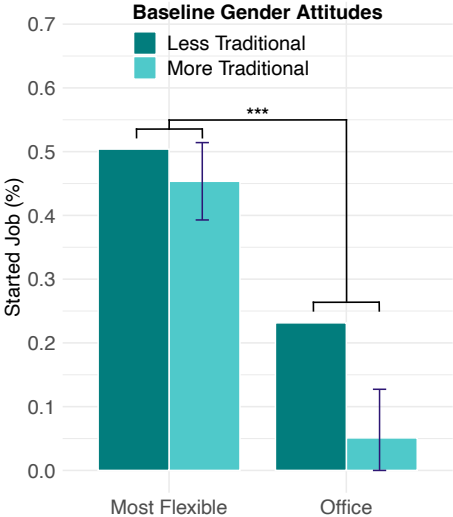
# Fact 1: traditional women's labor supply is more elastic with respect to flexibility



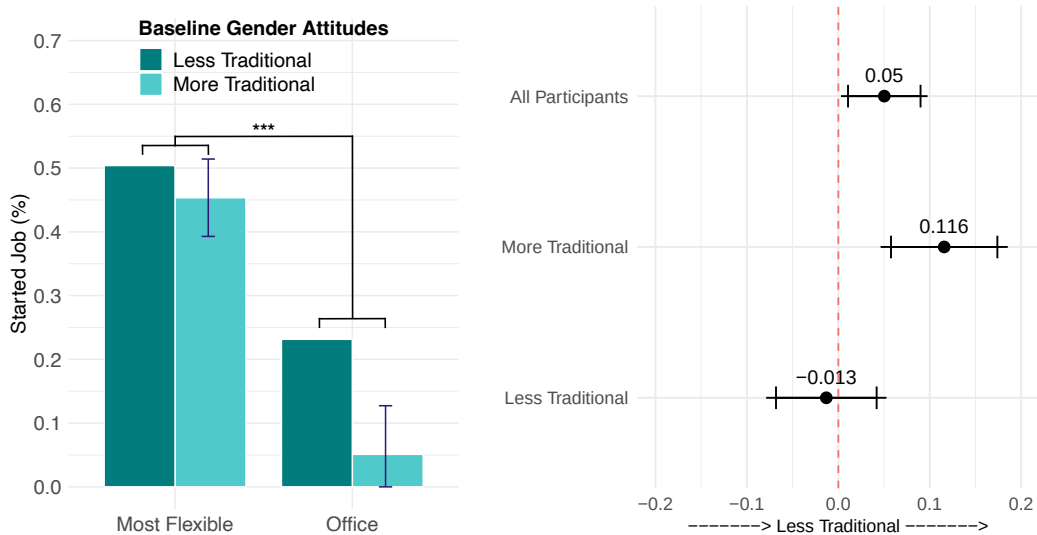
► Elasticity by other characteristics



# Fact 2: in turn, working for pay shifts attitudes to become less traditional

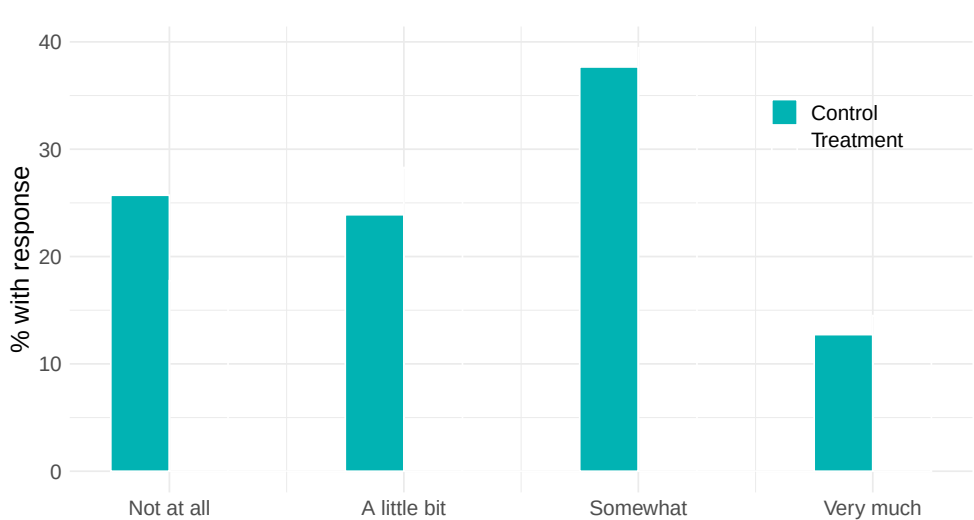


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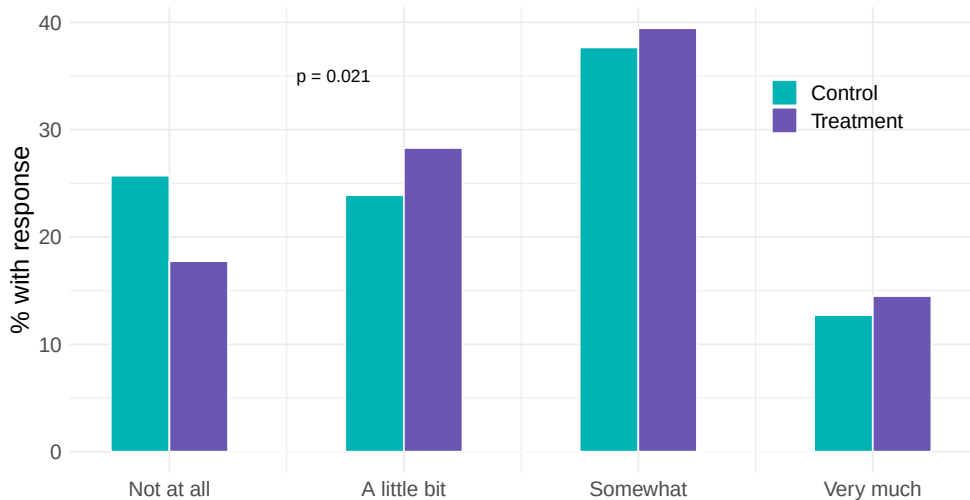


with the largest effects for women from more traditional households

Many participants believe that they are not using their full potential.



## Treatment increases belief that potential and talent are put to good use.



► Driven by women who worked

## Alternative mechanisms

### **Change in attitudes about appropriate behavior for women**

Workers marginal to work-from-home are from more traditional households, AND work experience makes women from more traditional households less traditional

▶ Not driven by ceiling effects   ▶ Heterogeneity by work arrangement   ▶ Index components   ▶ Definitions: attitudes vs beliefs

→ Learning about woman's abilities, level of household opposition, social image costs

1. ~~Learn skills or ability to signal skills to future employer~~ [~~shut down: we are the employer~~]
2. ~~Income effect~~ [~~would go in opposite direction~~]
3. ~~Increased bargaining power for woman earning income~~ [~~no effects on agency~~]
4. ~~Women's increased confidence~~ [~~can rule out small effects on self-efficacy~~]
5. Greater trust in employer

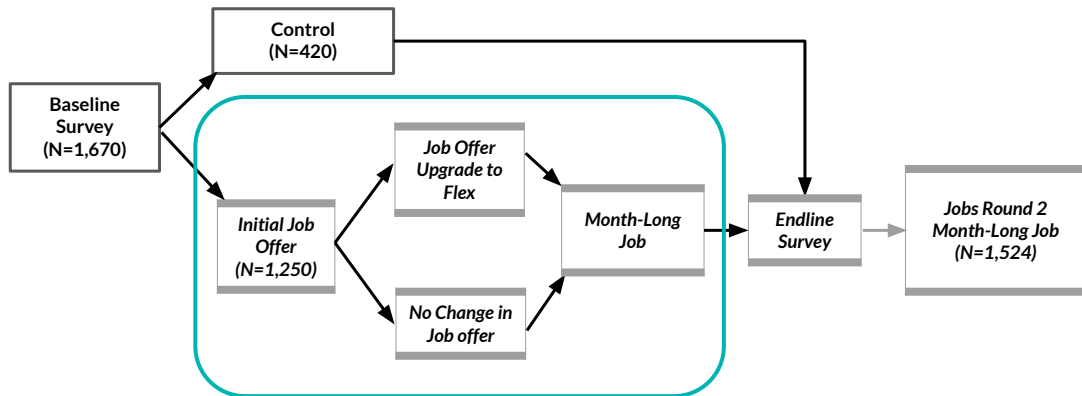
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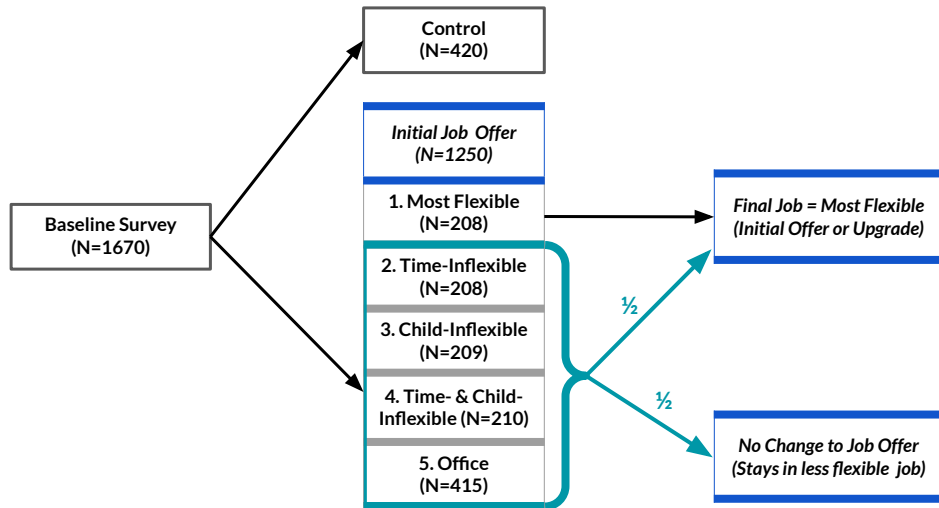
## Labor demand: effects of work-from-home on worker performance

For an employer considering introducing work-from-home...

1. What would the effects on performance be for workers previously working in person?
2. What new type of worker would job ads advertising work-from-home draw in to the firm?



## To separate selection from treatment, upgrade randomly selected participants



(Karlan & Zinman, 2009)

► Labor supply monotonic in flexibility



# Results: effects of work-from-home on job performance

## 1. Treatment (inframarginal workers):

- Work-from-home causes workers to do their tasks **20% more slowly** with **4% more errors**
  - Effects on accuracy
  - Effects on accuracy by task difficulty
  - Effects on speed
  - Effects on speed by task difficulty

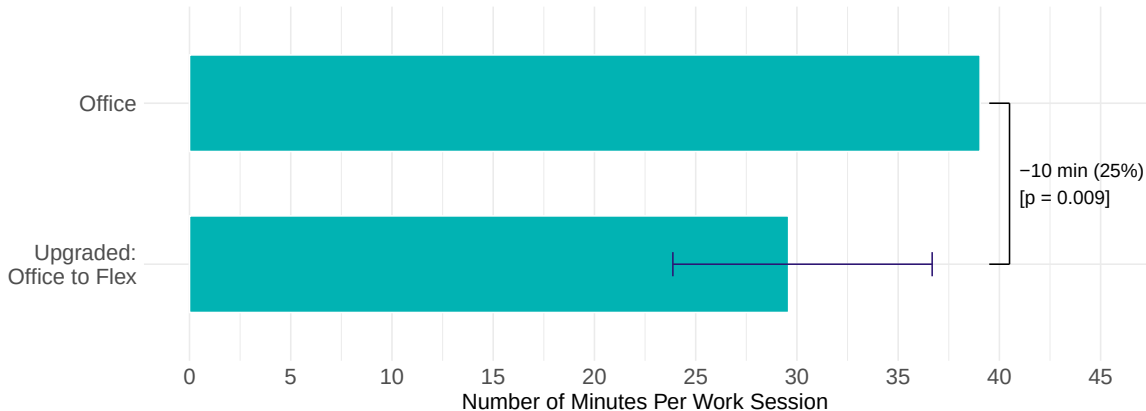
## 2. Selection (flexibility compliers):

- Workers whose labor supply is marginal to work-from-home not significantly different
  - Selection on accuracy
  - Selection on accuracy by task difficulty
  - Selection on speed
  - Selection on speed by task difficulty

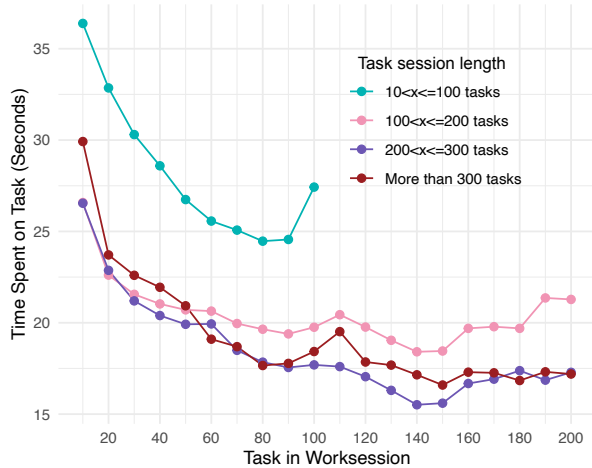
## Work-from-home reduces performance due to more fragmented work patterns.

**Work session:** a continuous segment in which  $<10$  mins passes between consecutive tasks

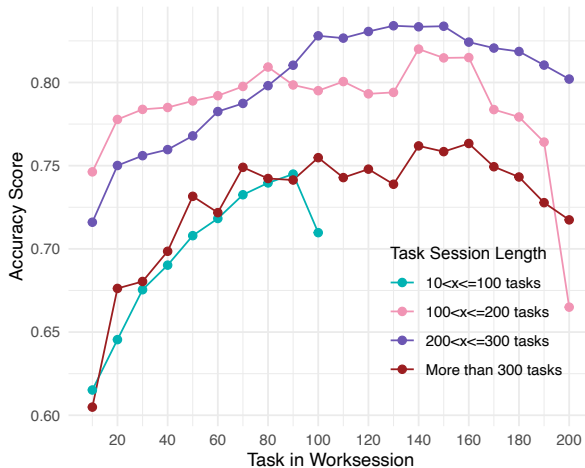
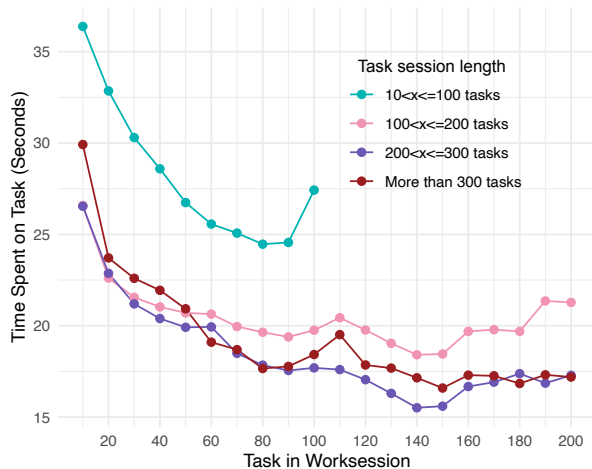
- **Result:** Work-from-home causes shorter work sessions



As workers get into the flow, they complete tasks more quickly



## As workers get into the flow, they complete tasks more quickly & more accurately



## Discussion

**[Takeaway]** flexible work arrangements are highly effective at increasing women's labor supply:

- Both immediately, and because some women will accept less flexible, outside-the-home work with a stepping stone but **they are not currently given the chance to take a gradual approach**

**[Implications for policy]** flexible arrangements may be underprovided for several reasons:

- **Lack of information:** e.g. employers may underestimate potential workforce they miss out on
- **Externalities:** benefits to women/other employers of providing gradual approach to broader labor market for out-of-labor-force women who want paid work

## Next steps: the dynamics of intermediate stops

### Questions

1. **Getting stuck:** Must the stepping stone be only *temporarily available* to be non-absorbing?
2. **Resistance:** Does taking many small steps or taking one big leap trigger more backlash?

### Study design:

- Application: transition from no paid work → part-time work → full-time work
- **Getting stuck:**
  - 1st round: randomize to no work vs part-time work
  - 2nd round: randomize to part-time, full-time, or choice of either
- **Resistance:**
  - Whether small steps vs large steps is better depends on relative size of (a) fixed cost of renegotiation vs (b) whether change is perceived to be threatening

# Demographics of study participants

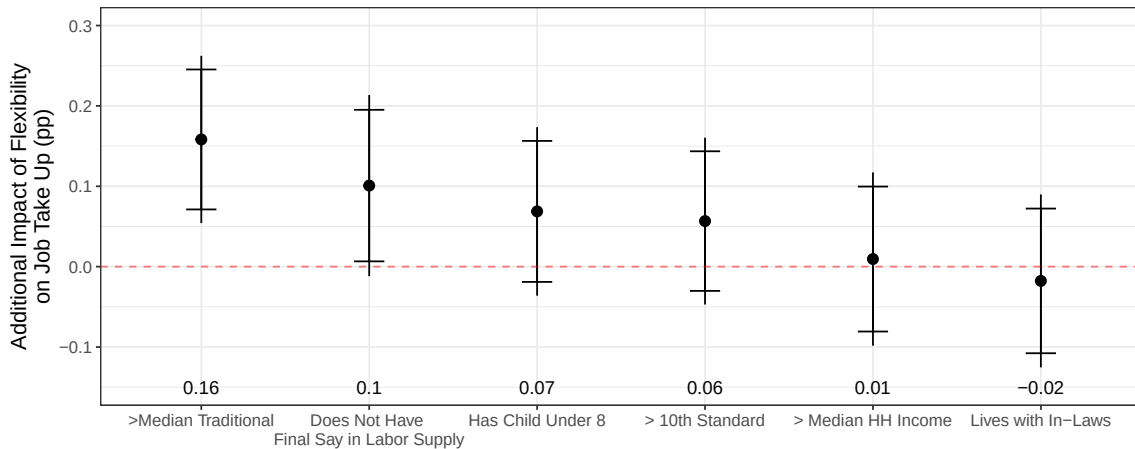
- **Average age:** 29.9 (SD: 8.2), nearly all (93%) married or previously married
- **Education:** 49% have finished 10th standard; 13% started undergrad degree
- **Household members:** avg HH size is 4.6 (SD: 1.7)
- **Religion & caste:** 39% are scheduled caste/scheduled tribe; 75% Hindu (25% Muslim)
- **Smartphone ownership:** 74% of participants own their own smartphone
- **Husbands' jobs:** 98% of husbands have worked in the last month  
(Common jobs: daily wage labor, private drivers, craftsmen, shipping, Swiggy/Ola/Zomato)

## Restrictiveness of inclusion criteria

1. Can read and speak Bangla or Hindi: 84% women in Kolkata (2011 census)
2. Smartphone access: 2 out of 3 rural West Bengal households have a smartphone (ASER 2022)
3. Labor force participation: 77% of women in West Bengal are out of the labor force (PLFS 2020)

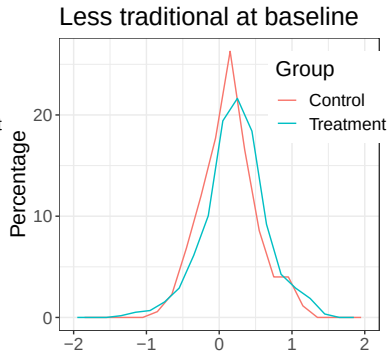
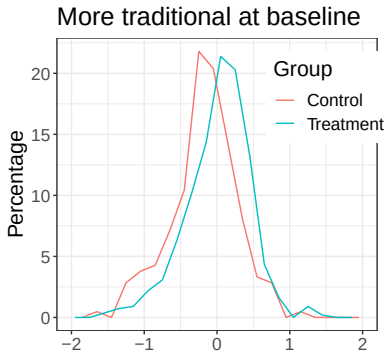
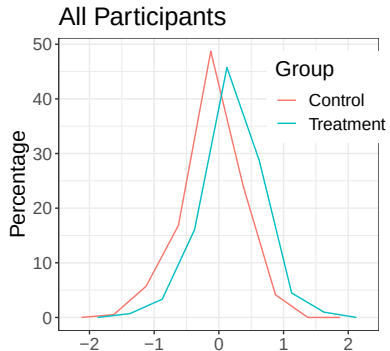


## Traditional attitudes generate largest labor supply response to flexibility



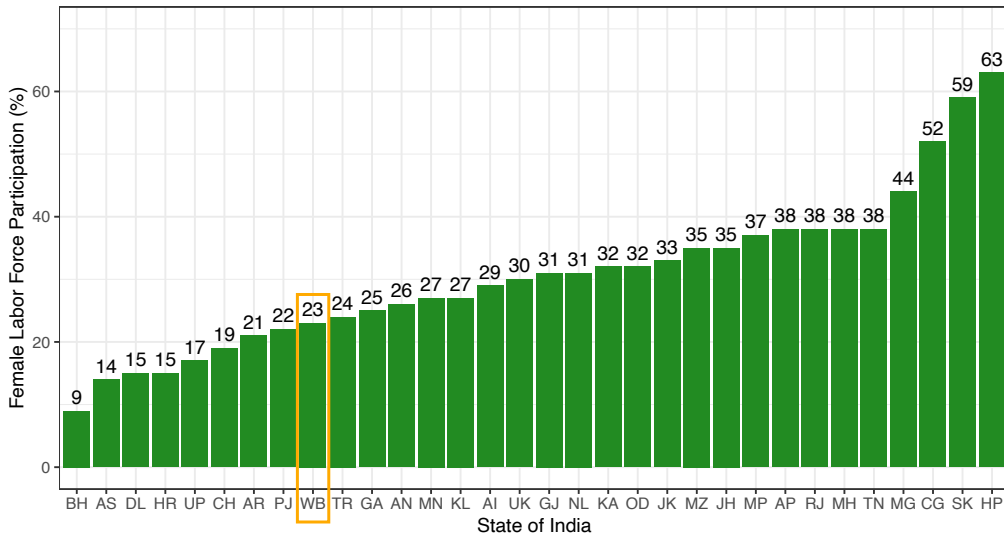
[◀ Back to elasticity by gender attitudes](#)

# Heterogeneity is not driven by ceiling effects

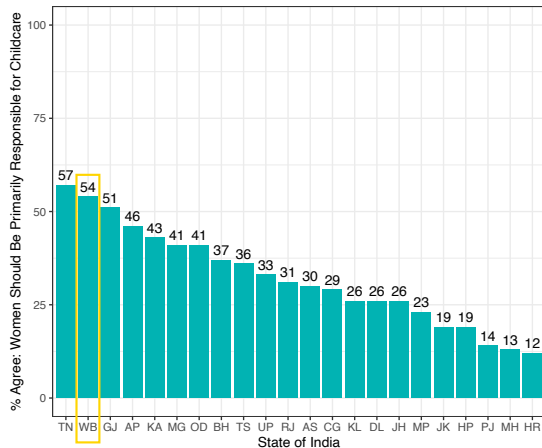
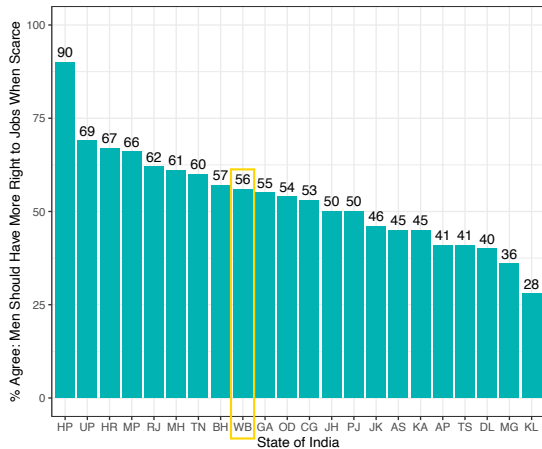


[◀ Back](#)

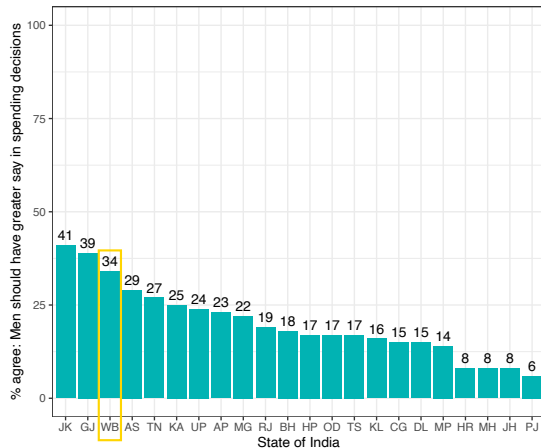
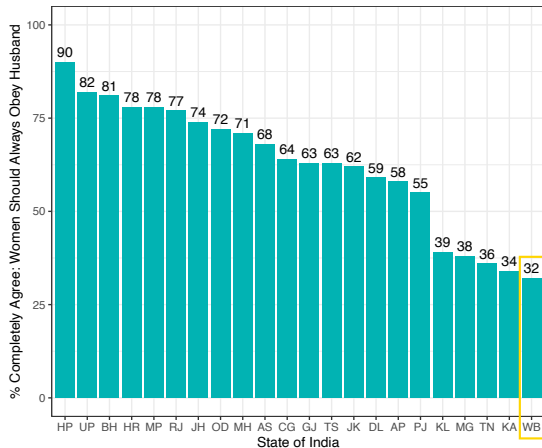
## West Bengal in Context - FI FP



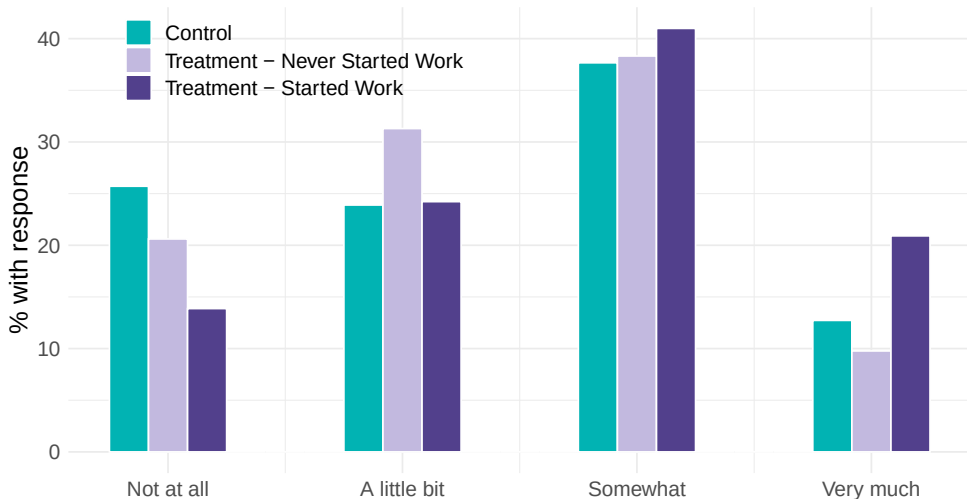
# West Bengal in Context - Attitudes about Gender Roles



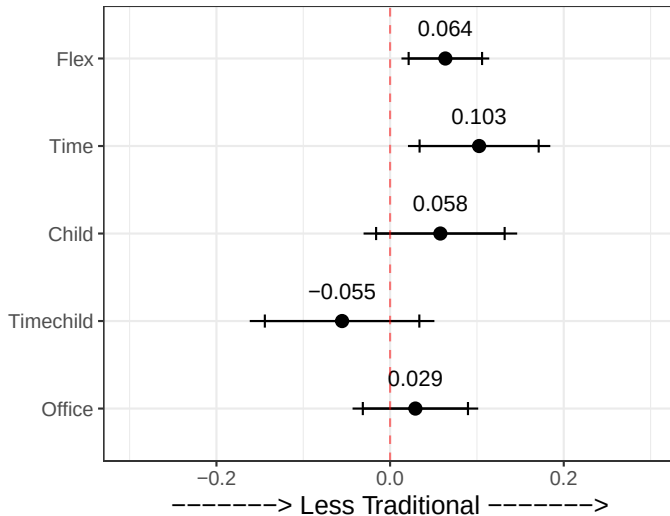
# West Bengal in Context - Attitudes about Decision Making



## Effect on using potential/talent is driven by women who worked.



## Treatment effects of work on attitudes: heterogeneity by work arrangement



## Non-digital work: Workers are sewing masks (left) and making bags (right)





## Measuring Gender Attitudes

- Measured as module of 16 (15) questions on baseline survey (endline survey)
- Participants asked to choose between hypothetical daughters- or sons-in-law
- Asked if they strongly disagree, disagree, agree, or strongly agree with statements such as:
  - “It is better for children if their mother does not have a job when they are young.”
  - “Husbands should split household responsibilities such as cooking, cleaning and childcare equally with their wives.”
- Index computed as a standardized, weighted average of these questions
  - Weights take into account the covariance structure of the components (Anderson, 2008)
  - Robustness checks with indices computed as simple averages produce similar results with less precise estimates

# Evidence from the endline survey that inflexible jobs are more costly to take up

Take up is  $\approx$  monotonically increasing (within person) in flexibility

| Did not accept job below, but accepted... | 1 less flexible job | 2 less flexible jobs | 3 less flexible jobs |
|-------------------------------------------|---------------------|----------------------|----------------------|
| Most Flexible                             | <1%                 | <1%                  | <1%                  |
| Time-Inflexible                           | 4.4%                | 1.7%                 | <1%                  |
| Child-Inflexible                          | 4.8%                | <1%                  | -                    |
| Timechild-Inflexible                      | 6.0%                | -                    | -                    |

## Balance table: endline

|                                       | All Participants | Control | Treatment | Pairwise t-test |
|---------------------------------------|------------------|---------|-----------|-----------------|
| Endline Complete (=1)                 | 0.913            | 0.919   | 0.911     | 0.585           |
| Age                                   | 29.955           | 29.640  | 30.062    | 0.232           |
| Completed 10th Standard (=1)          | 0.494            | 0.526   | 0.483     | 0.230           |
| Scheduled Caste/Tribe (=1)            | 0.395            | 0.394   | 0.396     | 0.995           |
| Hindu (=1)                            | 0.762            | 0.736   | 0.771     | 0.064*          |
| Never Married (=1)                    | 0.069            | 0.060   | 0.072     | 0.332           |
| Number HH Members                     | 4.603            | 4.687   | 4.575     | 0.254           |
| Parent-in-Law in HH (=1)              | 0.403            | 0.438   | 0.391     | 0.098*          |
| Has Child Under 8 (=1)                | 0.483            | 0.474   | 0.486     | 0.204           |
| Job Decision Final Say (=Self)        | 0.362            | 0.370   | 0.359     | 0.729           |
| Has Own Smartphone (=1)               | 0.725            | 0.731   | 0.723     | 0.642           |
| Gender Attitudes Index                | 0.029            | 0.001   | 0.038     | 0.102           |
| Agency Index                          | 0.017            | -0.005  | 0.025     | 0.441           |
| <i>Number of participants</i>         | 1525             | 386     | 1139      |                 |
| F-test of joint significance (F-stat) |                  |         |           | 1.251           |
| F-test, number of observations        |                  |         |           | 1525            |

## Balance table: randomization

|                                       | (1)<br>All Participants | (2)<br>Control | (3)<br>Treatment | (2)-(3)<br>Pairwise t-test |
|---------------------------------------|-------------------------|----------------|------------------|----------------------------|
| Age                                   | 29.911                  | 29.760         | 29.962           | 0.592                      |
| Completed 10th Standard (=1)          | 0.493                   | 0.510          | 0.488            | 0.617                      |
| Scheduled Caste/Tribe (=1)            | 0.385                   | 0.376          | 0.388            | 0.968                      |
| Hindu (=1)                            | 0.754                   | 0.724          | 0.764            | 0.094*                     |
| Never Married (=1)                    | 0.069                   | 0.062          | 0.071            | 0.298                      |
| Number HH Members                     | 4.626                   | 4.702          | 4.601            | 0.332                      |
| Parent-in-Law in HH (=1)              | 0.402                   | 0.433          | 0.392            | 0.119                      |
| Has Child Under 8 (=1)                | 0.478                   | 0.471          | 0.481            | 0.185                      |
| Job Decision Final Say (=Self)        | 0.359                   | 0.376          | 0.353            | 0.382                      |
| Has Own Smartphone (=1)               | 0.725                   | 0.731          | 0.723            | 0.366                      |
| Gender Attitudes Index                | 0.025                   | -0.000         | 0.034            | 0.112                      |
| Agency Index                          | 0.013                   | -0.000         | 0.018            | 0.674                      |
| <i>Number of participants</i>         | 1670                    | 420            | 1250             |                            |
| F-test of joint significance (F-stat) |                         |                |                  | 1.152                      |
| F-test, number of observations        |                         |                |                  | 1670                       |

## Work-from-home decreases accuracy by 4% (task-level regressions)

|                                         | Average task quality |                      |                      |                      |                      |                      |
|-----------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                         | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
| Upgraded to Work-from-Home              | -0.081***<br>(0.027) | -0.082***<br>(0.027) | -0.082***<br>(0.027) | -0.082***<br>(0.027) | -0.082***<br>(0.027) | -0.082***<br>(0.027) |
| Observations                            | 231,017              | 231,017              | 231,017              | 231,017              | 231,017              | 231,017              |
| Office Average Accuracy                 | 1.82                 | 1.82                 | 1.82                 | 1.82                 | 1.82                 | 1.82                 |
| Number Previous Tasks Completed         | ✓                    | ✓                    |                      |                      |                      |                      |
| Number Previous Tasks Completed Squared |                      | ✓                    |                      |                      |                      |                      |
| Previous Tasks Completed Fixed Effects  |                      |                      | ✓                    |                      |                      |                      |
| Number Previous Tasks Attempted         |                      |                      |                      | ✓                    | ✓                    |                      |
| Number Previous Tasks Attempted Squared |                      |                      |                      |                      | ✓                    |                      |
| Previous Tasks Attempted Fixed Effects  |                      |                      |                      |                      |                      | ✓                    |

Standard errors clustered at the participant level. [◀ Back to other impacts on productivity](#)

## Negative effect of work-from-home on accuracy is driven by difficult tasks

|                                             | Task Accuracy       |                     |                     |                     |                     |                     |
|---------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                             | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Upgraded to work-from-home                  | -0.022<br>(0.024)   | -0.023<br>(0.024)   | -0.025<br>(0.024)   | -0.023<br>(0.024)   | -0.024<br>(0.024)   | -0.028<br>(0.024)   |
| Difficult Task                              | -0.083**<br>(0.033) | -0.084**<br>(0.033) | -0.081**<br>(0.036) | -0.083**<br>(0.033) | -0.084**<br>(0.033) | -0.12***<br>(0.043) |
| Upgraded to work-from-home × Difficult Task | -0.20***<br>(0.054) | -0.20***<br>(0.054) | -0.19***<br>(0.055) | -0.20***<br>(0.054) | -0.20***<br>(0.054) | -0.18***<br>(0.053) |
| Observations                                | 231,017             | 231,017             | 231,017             | 231,017             | 231,017             | 231,017             |
| Office Workers Average Accuracy, Easy Task  | 1.84                | 1.84                | 1.84                | 1.84                | 1.84                | 1.84                |
| Number Previous Tasks Completed             | ✓                   | ✓                   |                     |                     |                     |                     |
| Number Previous Tasks Completed Squared     |                     | ✓                   |                     |                     |                     |                     |
| Previous Tasks Completed Fixed Effects      |                     |                     | ✓                   |                     |                     |                     |
| Number Previous Tasks Attempted             |                     |                     |                     | ✓                   | ✓                   |                     |
| Number Previous Tasks Attempted Squared     |                     |                     |                     |                     | ✓                   |                     |
| Previous Tasks Attempted Fixed Effects      |                     |                     |                     |                     |                     | ✓                   |

Standard errors clustered at the participant level. [◀ Back to other impacts on productivity](#)

## Work-from-home reduces worker speed by 20% (task-level regressions)

|                                         | Time spent per task |                   |                  |                  |                  |                  |
|-----------------------------------------|---------------------|-------------------|------------------|------------------|------------------|------------------|
|                                         | (1)                 | (2)               | (3)              | (4)              | (5)              | (6)              |
| Upgraded to Work-from-Home              | 5.09**<br>(2.11)    | 5.31***<br>(2.01) | 5.32**<br>(2.04) | 5.10**<br>(2.04) | 5.14**<br>(2.04) | 5.07**<br>(2.06) |
| Observations                            | 227,994             | 227,994           | 227,994          | 227,994          | 227,994          | 227,994          |
| Office Seconds Per Task                 | 25.1                | 25.1              | 25.1             | 25.1             | 25.1             | 25.1             |
| Number Previous Tasks Completed         | ✓                   | ✓                 |                  |                  |                  |                  |
| Number Previous Tasks Completed Squared |                     | ✓                 |                  |                  |                  |                  |
| Previous Tasks Completed Fixed Effects  |                     |                   | ✓                |                  |                  |                  |
| Number Previous Tasks Attempted         |                     |                   |                  | ✓                | ✓                |                  |
| Number Previous Tasks Attempted Squared |                     |                   |                  |                  | ✓                |                  |
| Previous Tasks Attempted Fixed Effects  |                     |                   |                  |                  |                  | ✓                |

Time spent on task measured as the time between task submissions, excluding the first task in a given worksession (i.e. when there is no task within the 10 minutes beforehand). Standard errors clustered at the participant level.

## Effects of work-from-home on speed, heterogeneity by task difficulty

|                                             | Time spent per task |                   |                   |                   |                   |                   |
|---------------------------------------------|---------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                             | (1)                 | (2)               | (3)               | (4)               | (5)               | (6)               |
| Upgraded to work-from-home                  | 3.35**<br>(1.59)    | 3.60**<br>(1.48)  | 3.67**<br>(1.51)  | 3.36**<br>(1.52)  | 3.42**<br>(1.51)  | 3.65**<br>(1.51)  |
| Difficult Task                              | 17.3***<br>(1.76)   | 17.5***<br>(1.77) | 17.2***<br>(1.79) | 17.3***<br>(1.76) | 17.5***<br>(1.76) | 18.1***<br>(2.15) |
| Upgraded to work-from-home × Difficult Task | 5.84**<br>(2.34)    | 5.85**<br>(2.35)  | 5.67**<br>(2.34)  | 5.85**<br>(2.34)  | 5.86**<br>(2.34)  | 5.05**<br>(2.35)  |
| Observations                                | 227,994             | 227,994           | 227,994           | 227,994           | 227,994           | 227,994           |
| Office Workers Average Time, Easy Task      | 20.3                | 20.3              | 20.3              | 20.3              | 20.3              | 20.3              |
| Number Previous Tasks Completed             | ✓                   | ✓                 |                   |                   |                   |                   |
| Number Previous Tasks Completed Squared     |                     | ✓                 |                   |                   |                   |                   |
| Previous Tasks Completed Fixed Effects      |                     |                   | ✓                 |                   |                   |                   |
| Number Previous Tasks Attempted             |                     |                   |                   | ✓                 | ✓                 |                   |
| Number Previous Tasks Attempted Squared     |                     |                   |                   |                   | ✓                 |                   |
| Previous Tasks Attempted Fixed Effects      |                     |                   |                   |                   |                   | ✓                 |

Standard errors clustered at the participant level. [◀ Back to other impacts on productivity](#)



## Work-from-home compliers are not significantly selected on accuracy

|                                         | Task Accuracy     |                   |                  |                  |                  |                  |
|-----------------------------------------|-------------------|-------------------|------------------|------------------|------------------|------------------|
|                                         | (1)               | (2)               | (3)              | (4)              | (5)              | (6)              |
| Work-from-home compliers                | -0.000<br>(0.029) | -0.001<br>(0.029) | 0.001<br>(0.030) | 0.000<br>(0.029) | 0.001<br>(0.029) | 0.001<br>(0.030) |
| Observations                            | 416,562           | 416,562           | 416,562          | 416,562          | 416,562          | 416,562          |
| Office Accepters Average Accuracy       | 1.71              | 1.71              | 1.71             | 1.71             | 1.71             | 1.71             |
| Number Previous Tasks Completed         | ✓                 | ✓                 |                  |                  |                  |                  |
| Number Previous Tasks Completed Squared |                   | ✓                 |                  |                  |                  |                  |
| Previous Tasks Completed Fixed Effects  |                   |                   | ✓                |                  |                  |                  |
| Number Previous Tasks Attempted         |                   |                   |                  | ✓                | ✓                |                  |
| Number Previous Tasks Attempted Squared |                   |                   |                  |                  | ✓                |                  |
| Previous Tasks Attempted Fixed Effects  |                   |                   |                  |                  |                  | ✓                |

Standard errors clustered at the participant level. [◀ Back to other impacts on productivity](#)

## Work-from-home compliers selection on accuracy by task difficulty

|                                              | Task Accuracy       |                     |                     |                     |                     |                     |
|----------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                              | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Work-from-home compliers                     | -0.007<br>(0.023)   | -0.005<br>(0.023)   | -0.005<br>(0.024)   | -0.005<br>(0.024)   | -0.006<br>(0.023)   | -0.006<br>(0.024)   |
| Difficult Task                               | -0.28***<br>(0.043) | -0.28***<br>(0.043) | -0.27***<br>(0.044) | -0.28***<br>(0.043) | -0.28***<br>(0.043) | -0.29***<br>(0.047) |
| Work-from-home compliers × Difficult Task    | 0.020<br>(0.056)    | 0.020<br>(0.056)    | 0.016<br>(0.057)    | 0.020<br>(0.056)    | 0.020<br>(0.056)    | 0.021<br>(0.056)    |
| Observations                                 | 416,562             | 416,562             | 416,562             | 416,562             | 416,562             | 416,562             |
| Office Accepters Average Accuracy, Easy Task | 1.79                | 1.79                | 1.79                | 1.79                | 1.79                | 1.79                |
| Number Previous Tasks Completed              | ✓                   | ✓                   |                     |                     |                     |                     |
| Number Previous Tasks Completed Squared      |                     | ✓                   |                     |                     |                     |                     |
| Previous Tasks Completed Fixed Effects       |                     |                     | ✓                   |                     |                     |                     |
| Number Previous Tasks Attempted              |                     |                     |                     | ✓                   | ✓                   |                     |
| Number Previous Tasks Attempted Squared      |                     |                     |                     |                     | ✓                   |                     |
| Previous Tasks Attempted Fixed Effects       |                     |                     |                     |                     |                     | ✓                   |

Standard errors clustered at the participant level. [◀ Back to other impacts on productivity](#)

## Work-from-home compliers: selection on speed

|                                         | Time spent per task |                |                |                |                |                |
|-----------------------------------------|---------------------|----------------|----------------|----------------|----------------|----------------|
|                                         | (1)                 | (2)            | (3)            | (4)            | (5)            | (6)            |
| Work-from-home compliers                | 2.04<br>(1.98)      | 1.59<br>(1.79) | 1.49<br>(1.77) | 1.92<br>(1.95) | 2.04<br>(1.95) | 2.07<br>(1.97) |
| Observations                            | 409,300             | 409,300        | 409,300        | 409,300        | 409,300        | 409,300        |
| Office Accepters Average Seconds        | 27.9                | 27.9           | 27.9           | 27.9           | 27.9           | 27.9           |
| Number Previous Tasks Completed         | ✓                   | ✓              |                |                |                |                |
| Number Previous Tasks Completed Squared |                     | ✓              |                |                |                |                |
| Previous Tasks Completed Fixed Effects  |                     |                | ✓              |                |                |                |
| Number Previous Tasks Attempted         |                     |                |                | ✓              | ✓              |                |
| Number Previous Tasks Attempted Squared |                     |                |                |                | ✓              |                |
| Previous Tasks Attempted Fixed Effects  |                     |                |                |                |                | ✓              |

Standard errors clustered at the participant level. [◀ Back to other impacts on productivity](#)

## Work-from-home compliers: selection on speed by task difficulty

|                                              | Time spent per task |                   |                   |                   |                   |                   |
|----------------------------------------------|---------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                              | (1)                 | (2)               | (3)               | (4)               | (5)               | (6)               |
| Work-from-home compliers                     | 2.07<br>(1.58)      | 1.59<br>(1.40)    | 1.56<br>(1.40)    | 1.94<br>(1.55)    | 2.08<br>(1.55)    | 1.99<br>(1.55)    |
| Difficult Task                               | 23.1***<br>(1.53)   | 23.3***<br>(1.53) | 22.7***<br>(1.54) | 23.2***<br>(1.53) | 23.4***<br>(1.53) | 22.9***<br>(1.63) |
| Work-from-home compliers × Difficult Task    | 0.057<br>(2.10)     | 0.094<br>(2.11)   | -0.077<br>(2.05)  | 0.069<br>(2.10)   | 0.092<br>(2.10)   | 0.273<br>(2.09)   |
| Observations                                 | 409,300             | 409,300           | 409,300           | 409,300           | 409,300           | 409,300           |
| Office Accepters Average Accuracy, Easy Task | 21.4                | 21.4              | 21.4              | 21.4              | 21.4              | 21.4              |
| Number Previous Tasks Completed              | ✓                   | ✓                 |                   |                   |                   |                   |
| Number Previous Tasks Completed Squared      |                     | ✓                 |                   |                   |                   |                   |
| Previous Tasks Completed Fixed Effects       |                     |                   | ✓                 |                   |                   |                   |
| Number Previous Tasks Attempted              |                     |                   |                   | ✓                 | ✓                 |                   |
| Number Previous Tasks Attempted Squared      |                     |                   |                   |                   | ✓                 |                   |
| Previous Tasks Attempted Fixed Effects       |                     |                   |                   |                   |                   | ✓                 |

Standard errors clustered at the participant level. [◀ Back to other impacts on productivity](#)

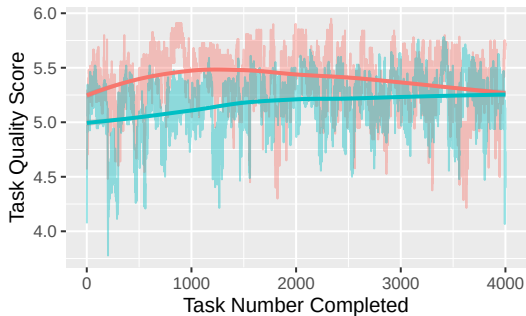
## No significant difference in take up of digital vs non-digital jobs

|                                          | Started Work in Round 2 |                 |
|------------------------------------------|-------------------------|-----------------|
| Round 2 Job = Digital                    | 0.02<br>(0.03)          | 0.05<br>(0.06)  |
| Treatment                                |                         | 0.03<br>(0.06)  |
| Round 2 Job = Digital $\times$ Treatment |                         | -0.03<br>(0.07) |
| Double Post Lasso Controls + Strata FE   | ✓                       | ✓               |
| Observations                             | 1,524                   | 1,524           |

[◀ Back to round 2 diagram](#)

## Quality by tasks completed so far.

Left: All workers



Work Arrangement — Office — Upgraded to WFH

Right: Workers who completed  $> 90\%$  of tasks (3600)



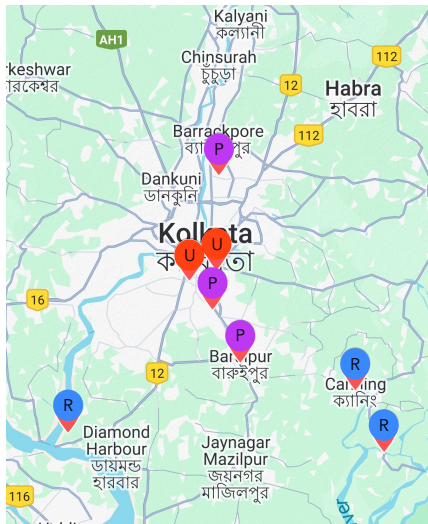
Work Arrangement — Office — Upgraded to WFH

[◀ Back to impacts on productivity](#)

## Definitions: Beliefs vs Attitudes vs Norms

- **Beliefs:** an individual's estimate of a value that can be correct or incorrect
- **Attitudes:** an individual's own preferences/opinions that together make up personal values – cannot be correct or incorrect
- **Norms:** informal rules that most people accept & follow – applies to a group, not an individual
  - Descriptive norms: what people do
  - Injunctive norms: what people think others in the group believe they *should* do

## Map of study areas



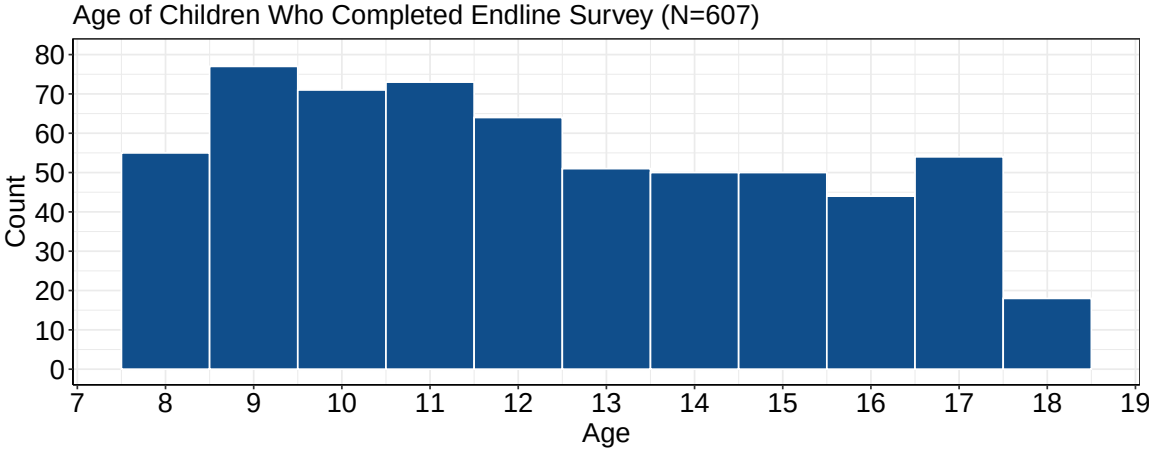
From west to east:

1. Noorpur (Rural)
2. New Alipore (Urban)
3. Atabagan (Peri-urban)
4. Tiljala (Urban)
5. Sodepur (Peri-urban)
6. Baruipur (Peri-urban)
7. Canning (Rural)
8. Basanti (Rural)

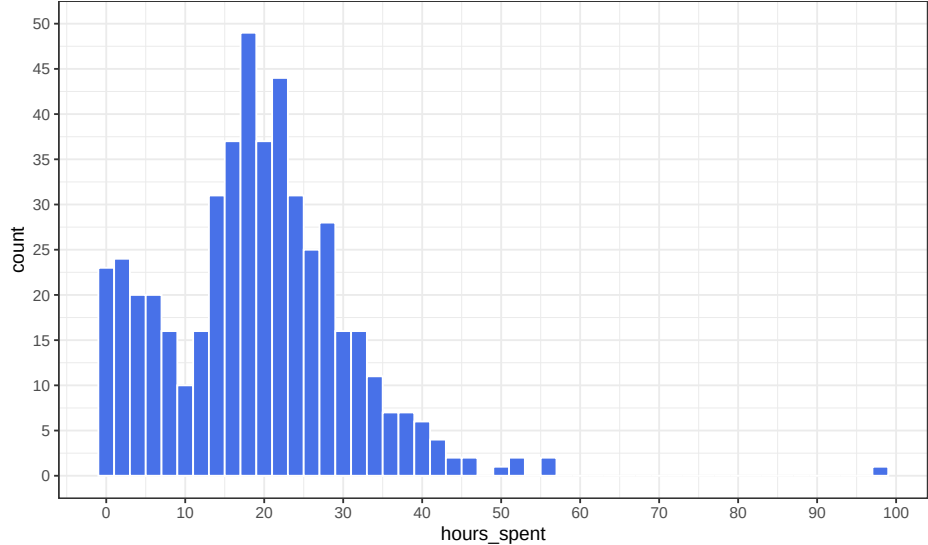
◀ [Back to Study Context](#)



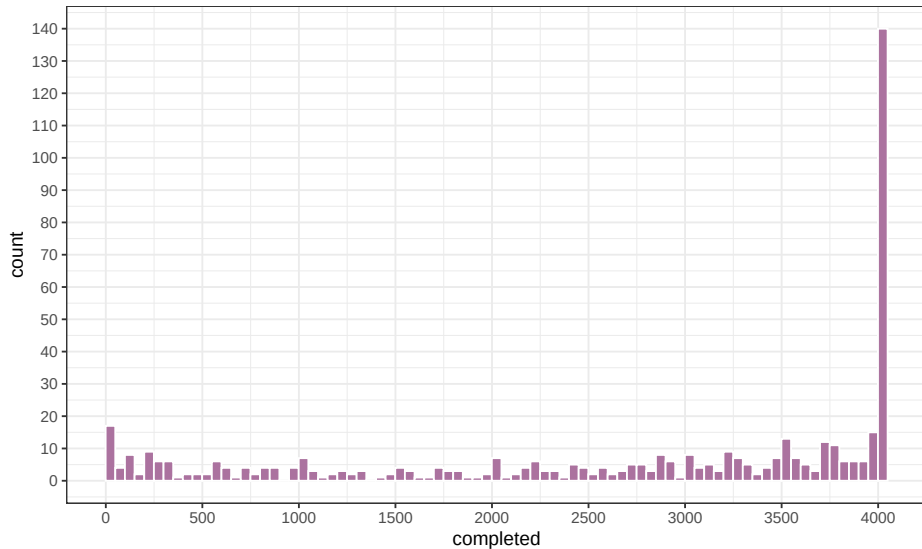
# Age of children surveyed



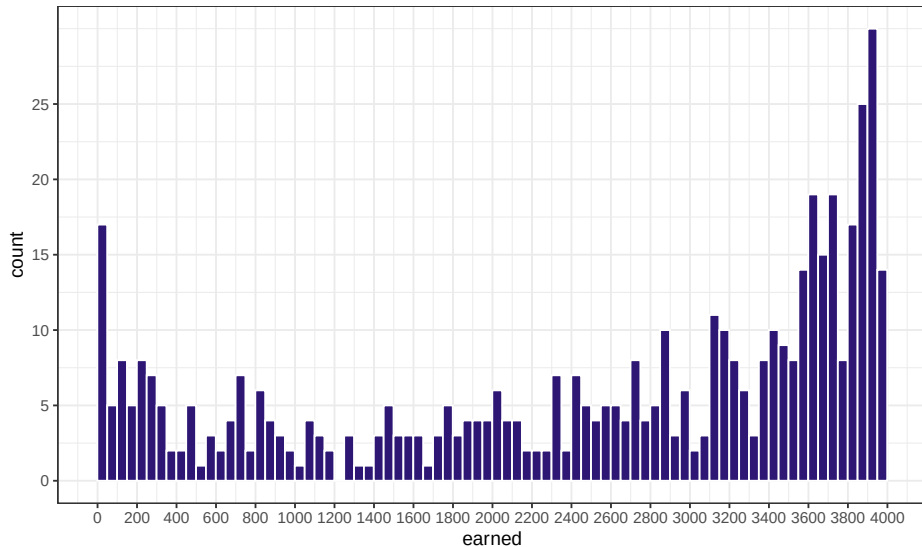
# Number of hours spent working



## Number of completed tasks submitted

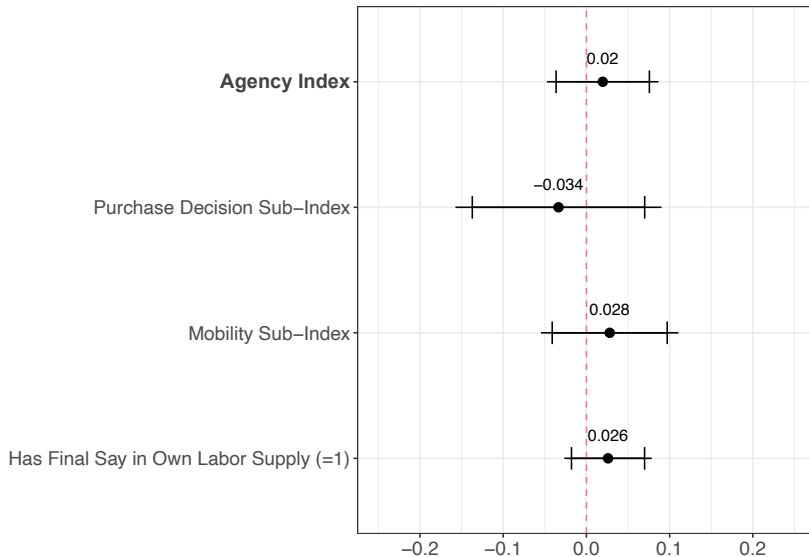


## Distribution of earnings



## Effect on gender attitudes index by components

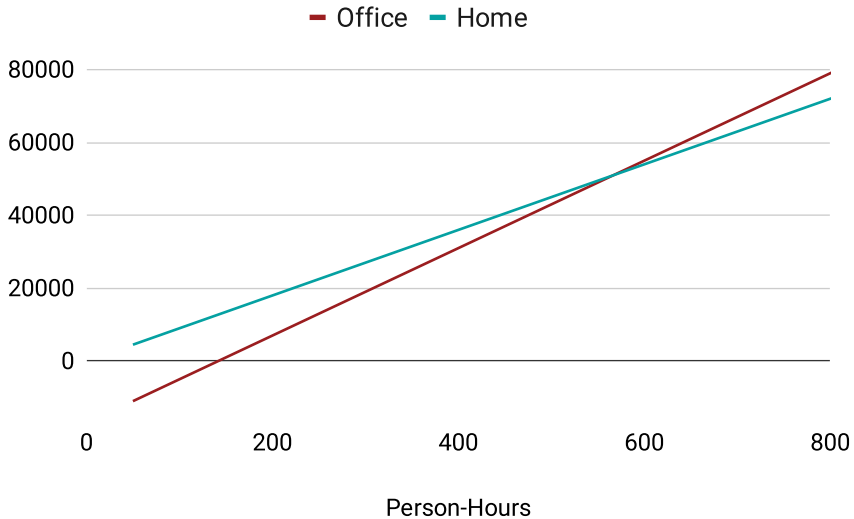
|                         | All Gender Attitudes<br>(1) | Women & Work<br>(2) | Tech<br>(3)     | Ability<br>(4)  | HH Roles<br>(5) |
|-------------------------|-----------------------------|---------------------|-----------------|-----------------|-----------------|
| Arrangement = Flex      | 0.06**<br>(0.03)            | 0.12***<br>(0.04)   | 0.05<br>(0.04)  | 0.03<br>(0.06)  | 0.03<br>(0.04)  |
| Arrangement = Time      | 0.10**<br>(0.04)            | 0.14**<br>(0.06)    | 0.06<br>(0.07)  | 0.10<br>(0.11)  | 0.05<br>(0.06)  |
| Arrangement = Child     | 0.07<br>(0.05)              | 0.12*<br>(0.07)     | 0.07<br>(0.07)  | -0.12<br>(0.11) | 0.10<br>(0.08)  |
| Arrangement = Timechild | -0.04<br>(0.06)             | -0.08<br>(0.08)     | -0.04<br>(0.08) | 0.00<br>(0.11)  | 0.04<br>(0.07)  |
| Arrangement = Office    | 0.03<br>(0.04)              | 0.02<br>(0.05)      | 0.07<br>(0.06)  | 0.12<br>(0.08)  | 0.03<br>(0.05)  |
| Observations            | 1,525                       | 1,524               | 1,525           | 968             | 1,525           |



## Cost vs productivity tradeoff assumptions

| Item                       | Value | Units    |
|----------------------------|-------|----------|
| Rent                       | 8000  | Rs/month |
| Office Supervisor          | 1500  | Rs/month |
| Other office materials     | 500   | Rs/month |
| Output quality (from home) | 4%    | Lower    |
| Speed (from home)          | 21%   | Slower   |
| Market wage                | 120   | Rs/hour  |

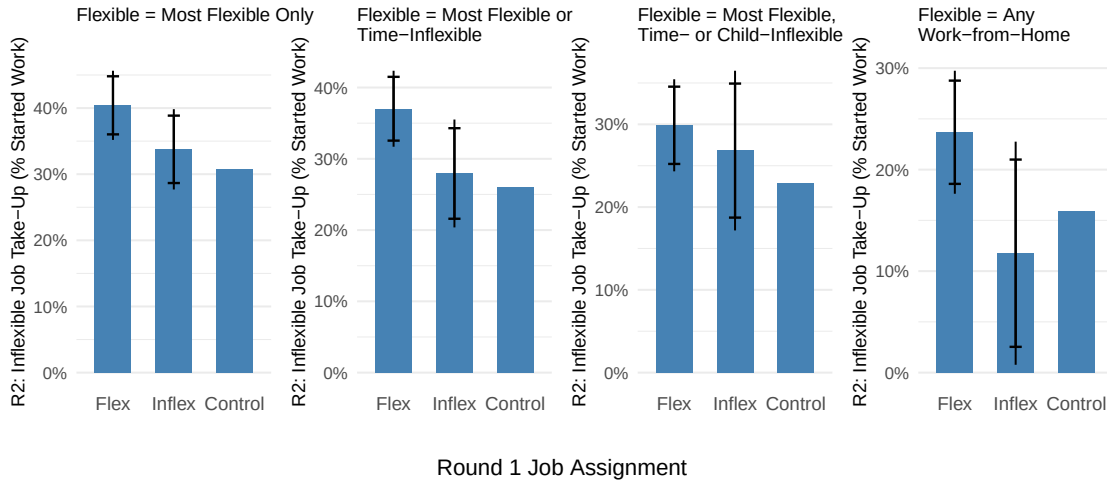
## Office vs Home Tradeoff





# Take up of Inflexible Round 2 Jobs, by Flexibility of Round 1 Job Assignment

## Never Had Paid Work Experience



# Take Up of Inflexible Round 2 Jobs, by Flexibility of Round 1 Job Assignment

## Has Had Paid Work Experience

